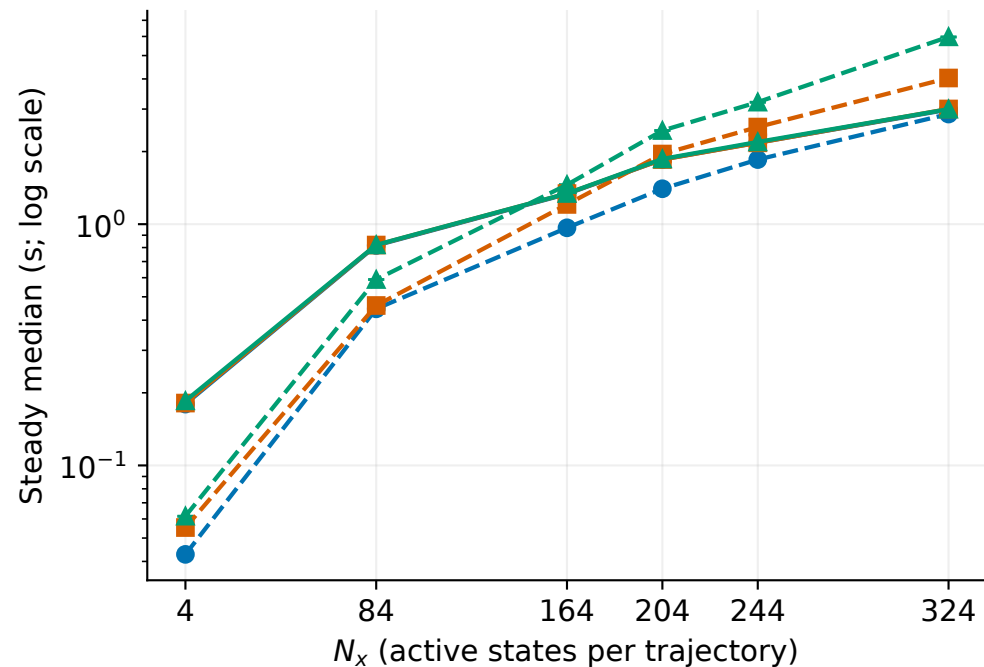


HH steady loss + gradient performance

H200 NVL | optimized HH | B=S=16, T=1600 | float64 | four batches, one worker per method/config

● Leak: $N_\theta = C = N_x/4$ ▲ Leak+K+Na: $N_\theta = 3C = 3N_x/4$ - - RTRL
■ Leak+K: $N_\theta = 2C = N_x/2$ — BPTT



Error bars: within-worker Q25-Q75 (often smaller than markers). $r < 1$ favors RTRL.
Lines are guides between measured points; no uncrossed boundary is asserted.

